# 화학반응속도론 (2014년 2학기)
# 제4장 광반응의 반응메커니즘

중요 학습 내용

- 광물리 과정과 광화학 과정
- 빛의 흡수와 방출, 복사 과정과 비복사 과정
- Jablonski 도표
- 형광 양자수득률과 형광 수명
- Stern-Volmer 도표
- 광감각제의 반응메커니즘
- FRET 반응의 반응메커니즘
- 광합성의 반응메커니즘



## 4.1 광화학

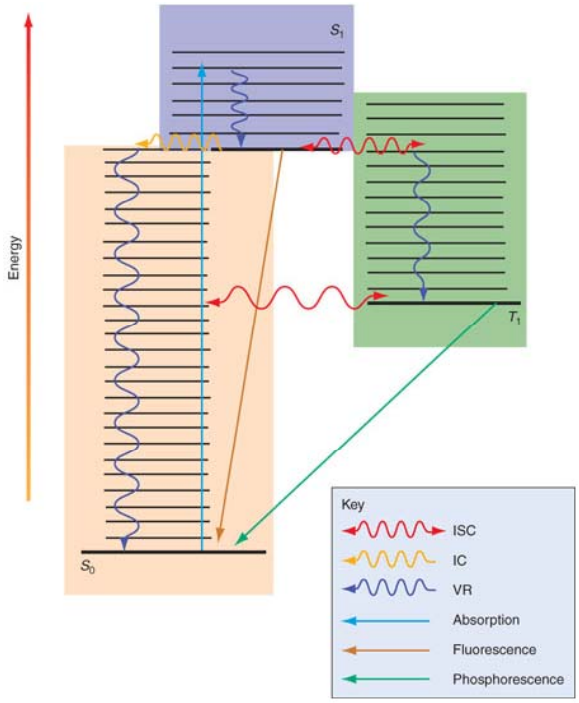
- 광화학 (photochemistry): 빛, 즉 전자기파의 흡수에 의해 개시되는 반응을 다루는 화학
- 광물리 과정 (photophysical process): 빛을 흡수하는 분자의 구조를 유지하면서 일어나는 과정.
- 광화학 과정 (photochemical process): 빛을 흡수하는 분자의 구조 변화 또는 원자나 전자의 이동이 수반되는 과정.

# 광물리 과정

1. 빛의 흡수 (absorption)에 의한 전자 들뜸
   1) $S_0 + h\nu \rightarrow S_1$: 일반적인 분자에 의한 빛의 흡수, $10^{-16}$-$10^{-15}$ s 이내에 발생
   2) $S_0 + h\nu \rightarrow T_1$: $S_0 \rightarrow S_1$ 흡수보다 흡수하는 빛의 세기가 $10^8$ 배 작음. 일반적으로 $S_0 \rightarrow T_1$ 흡수는 무시함.
2. 빛의 방출 (emission)에 의한 전자 이완: 복사과정 (radiative process)
   1) 형광 (fluorescence): $S_1 \rightarrow S_0 + h\nu$, $10^{-12}$-$10^{-6}$ s
   2) 인광 (phosphorescence): $T_1 \rightarrow S_0 + h\nu$ , 형광에 비해 매우 느리게 일어남. $10^{-6}$-$10^{-1}$ s
3. 빛의 방출이 없는 전자 이완: 비복사 과정 (nonradiative proces)
   1) 내부전환 (internal conversion, IC): $S_1 \rightarrow S_0$. $S_1$ 상태에 있는 전자가 빛을 방출하지 않고 원래의 $S_0$ 상태로 전이되는 과정. $10^{-14}$-$10^{-11}$ s
   2) 계간 교차 (intersystem crossing, ISC): $S_1 \rightarrow T_1$ 또는 $T_1 \rightarrow S_0$. $S_1$ 상태에 있는 전자가 스핀-궤도 상호작용에 의해 $T_1$ 상태로 전이되는 과정. 형광이나 내부전환에 비해 상대적으로 느리게 일어남. $10^{-12}$-$10^{-4}$ s

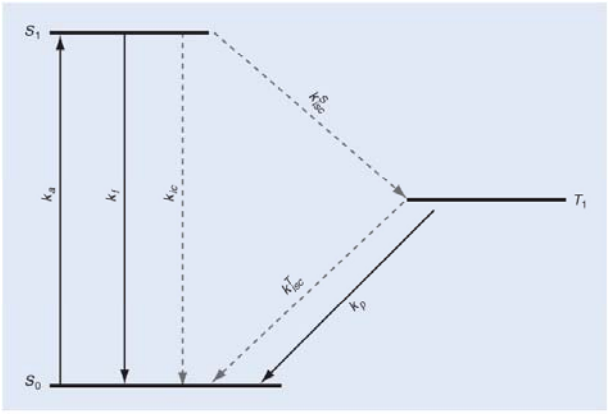


# Jablonski 도표



# 광물리 과정의 반응속도

- 광물리과정과 반응속도




| Process | Reaction | Rate |
|---|---|---|
| Absorption/excitation | $S_0 + h\nu \longrightarrow S_1$ | $k_a[S_0]$ $(k_a = I_0\sigma_A)$ |
| Fluorescence | $S_1 \longrightarrow S_0 + h\nu$ | $k_f[S_1]$ |
| Internal conversion | $S_1 \longrightarrow S_0$ | $k_{ic}[S_1]$ |
| Intersystem crossing | $S_1 \longrightarrow T_1$ | $k_{isc}^S[S_1]$ |
| Phosphorescence | $T_1 \longrightarrow S_0 + h\nu$ | $k_p[T_1]$ |
| Intersystem crossing | $T_1 \longrightarrow S_0$ | $k_{isc}^T[T_1]$ |



# 연속 광원과 형광 양자수득률

- 연속 광원: 묽은 농도의 시료에 강한 세기의 빛을 계속해서 조사할 때 사용되는 광원.
- 연속 광원 하에서 광물리 과정의 속도법칙

1) $S_0$ 상태의 속도법칙:

$$\frac{d[S_0]}{dt} = -k_a[S_0] \propto I_{abs}$$

2) $S_1$ 상태의 속도법칙:

$$\frac{d[S_1]}{dt} = k_a[S_0] - k_{ic}[S_1] - k_{isc}[S_1] - k_f[S_1]$$

3) $T_1$ 상태의 속도법칙:

$$\frac{d[T_1]}{dt} = k_{isc}[S_1] - k_p[T_1]$$

묽은 시료 하에서 $S_1$ 상태의 시료 양은 매우 적고, 그 변화량이 매우 작으므로 정류상태근사법을 사용

$$\frac{d[S_1]}{dt} = k_a[S_0] - k_{ic}[S_1] - k_{isc}[S_1] - k_f[S_1] \approx 0$$

$$\Rightarrow k_a[S_0] = (k_{ic} + k_{isc} + k_f)[S_1]$$

- 형광 양자수득률 (fluorescence quantum yield):

$$\Phi_f = \frac{(\text{단위 시간 당}) \text{형광에 의해 방출한 광자 수}}{(\text{단위 시간 당}) \text{흡수한 광자 수}}$$

$$\Phi_f = \frac{k_f[S_1]}{k_a[S_0]} = \frac{k_f}{k_f + k_{ic} + k_{isc}} \qquad (0 \le \Phi_f \le 1)$$

| Compound | $\Phi_F$[a] | $\varepsilon_{max}$ | $k_F^0$ |
|---|---|---|---|
| Benzene | ~0.2 | 250 | $2 \times 10^6$ |
| Naphthalene | ~0.2 | 270 | $2 \times 10^6$ |
| Anthracene | ~0.4 | 8500 | $5 \times 10^7$ |
| Tetracene | ~0.2 | 14000 | $2 \times 10^7$ |
| 9,10-Diphenylanthracene | ~1.0 | 12600 | $\sim 5 \times 10^8$ |
| Pyrene | ~0.7 | 510 | $\sim 10^6$ |
| Triphenylene | ~0.1 | 355 | $\sim 2 \times 10^6$ |
| Perylene | ~1.0 | 39500 | $\sim 10^8$ |
| Stilbene[b] | ~0.05 | 24000 | $\sim 10^8$ |
| 1-Chloronaphthalene | ~0.05 | ~300 | $\sim 10^6$ |
| 1-Bromonaphthalene | ~0.002 | ~300 | $\sim 10^6$ |
| 1-Iodonaphthalene | ~0.000 | ~300 | $\sim 10^6$ |
| Benzophenone[c] | ~0.000 | ~200 | $\sim 10^6$ |
| Biacetyl[d] | ~0.002 | ~20 | $\sim 10^5$ |
| Diaza[2.2.2]bicyclooctane[e] | ~1.0 | ~200 | $\sim 10^6$ |
| Acetone[f] | ~0.001 | ~20 | $\sim 10^5$ |
| Perfluoroacetone[g] | ~0.1 | ~20 | $\sim 10^5$ |
| 3-Bromoperylene[h] | ~1.0 | ~40,000 | $\sim 10^8$ |
| Pyrene-3-carboxaldehyde[i] | ~0.25 | ~70,000 | $\sim 10^8$ |
| Cyclobutanone[j] | ~0.0001 | ~20 | $\sim 10^5$ |
| Diaza[2.2.1]bicycloheptane[e] | ~0.0001 | 400 | $\sim 10^6$ |



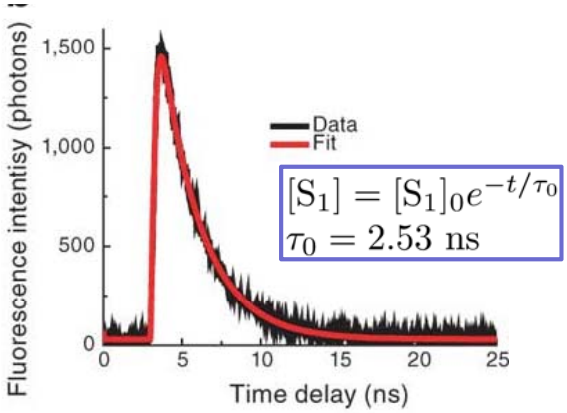
## 펄스 광원과 형광 수명

- 펄스 광원: 시료에 광원을 순간적으로만 조사할 때 사용하는 광원
- 펄스 광원 하에서 $S_1$ 상태의 속도법칙

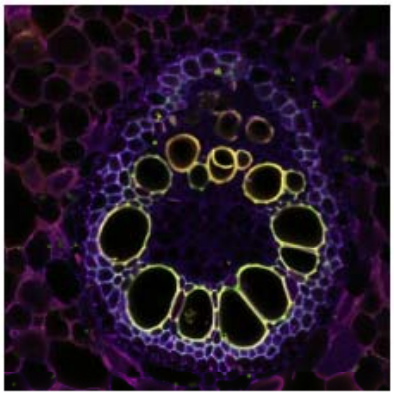
$$\frac{d[S_1]}{dt} = -k_f[S_1] - k_{ic}[S_1] - k_{isc}[S_1]$$
$$= -(k_f + k_{ic} + k_{isc})[S_1]$$
$$= -\frac{1}{\tau_0}[S_1] \qquad \left(\frac{1}{\tau_0} = (k_f + k_{ic} + k_{isc})\right)$$

$$[S_1] = [S_1]_0 e^{-t/\tau_0}$$ ($\tau_0$: 형광수명)

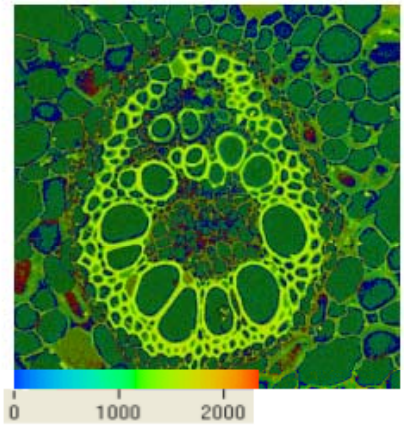
$$\Phi_f = k_f \tau_0$$



# Fluorescence life time imaging microscopy (FLIM)



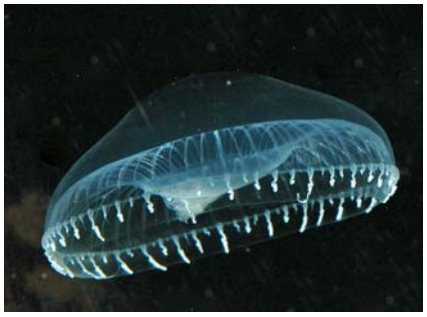
Conventional confocal intensity image



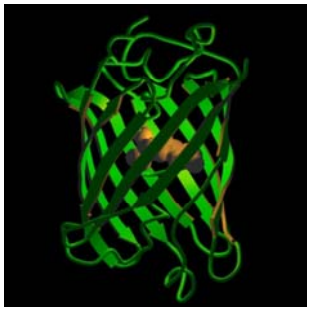

Fluorescence lifetime image /ps



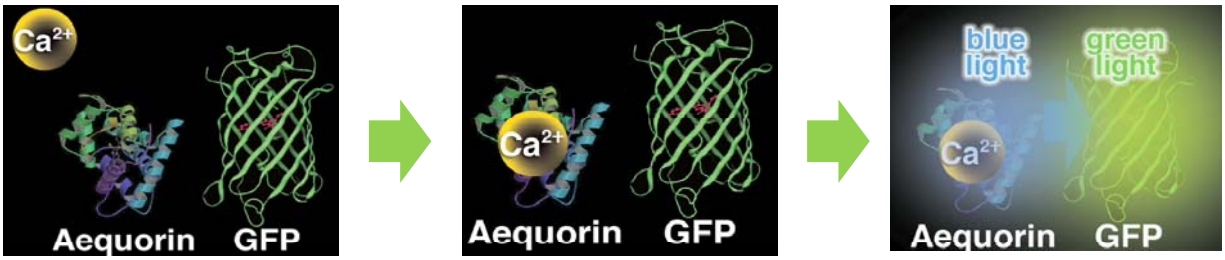
# What is Green Fluorescent Protein (GFP)?



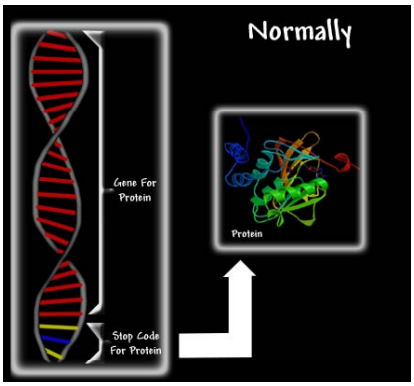
Aequorea victoria jellyfish

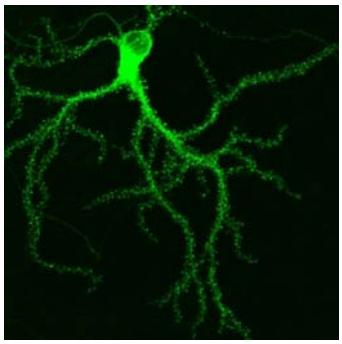


GFP

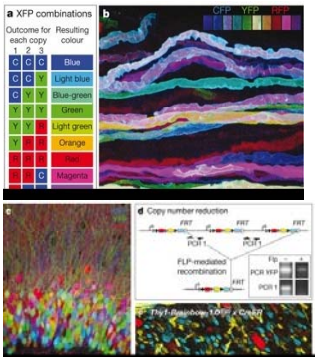


# Applications of GFP

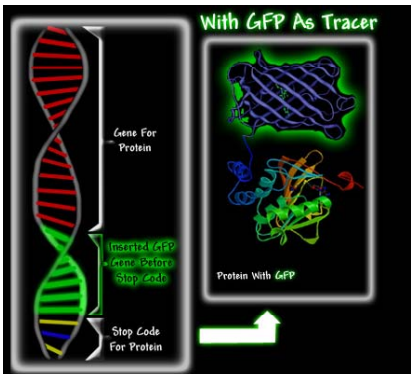


Neuron GFP

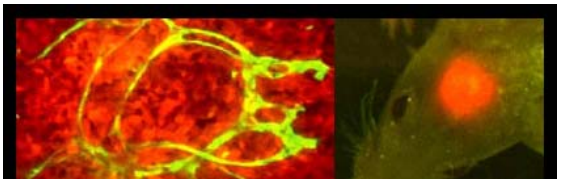



Livet et al., Nature **450**, 56 (2007)






GFP cloned 쥐와 보통 쥐



암세포들 (붉은색)과 혈관 (녹색)

http://www.conncoll.edu/ccacad/zimmer/GFP-ww/GFP-1.htm

## The Nobel Prize in Chemistry 2008

"for the discovery and development of the green fluorescent protein, GFP"

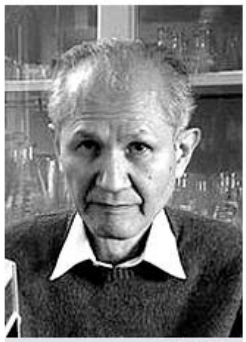


Photo: J. Henriksson/SCANPIX

**Osamu Shimomura**

1/3 of the prize

USA

Marine Biological Laboratory (MBL)
Woods Hole, MA, USA

b. 1928



Photo: J. Henriksson/SCANPIX

**Martin Chalfie**

1/3 of the prize

USA

Columbia University
New York, NY, USA

b. 1947



Photo: UCSD

**Roger Y. Tsien**

1/3 of the prize

USA

University of California
San Diego, CA, USA

b. 1952

# 4.2 형광-소광 반응과 Stern-Volmer 식

- 소광 (quenching): 분자 간 충돌에 의한 비활성화, 에너지 전달 또는 전자전달 등에 의해 전자의 들뜬 상태의 수명이 줄어드는 것. 소광에 의해 형광 양자수득률이 감소한다.
- Stern-Volmer 식 방법 ($k_{ic}$와 $k_{isc}$의 값은 매우 작아 무시한다고 가정)

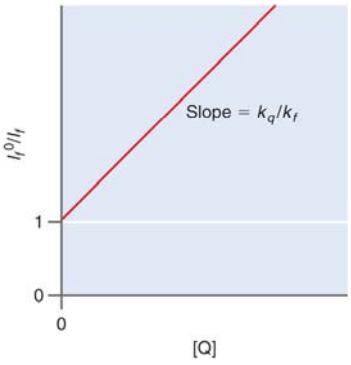
1) 연속 광원에 의한 빛의 흡수: $S_0 + h\nu \xrightarrow{k_a} S_1$

2) 형광 반응: $S_1 \xrightarrow{k_f} S_0 + h\nu$

3) 소광 반응: $S_1 + Q \xrightarrow{k_q} S_0 + Q$

$$\frac{d[S_1]}{dt} = k_a[S_0] - k_f[S_1] - k_q[Q][S_1] \approx 0$$

$$\Rightarrow [S_1] = \frac{k_a[S_0]}{k_f + k_q[Q]}$$




$$I_f = k_f[S_1] = \frac{k_f k_a[S_0]}{k_f + k_q[Q]} \qquad \Rightarrow \frac{1}{I_f} = \frac{k_f + k_q[Q]}{k_f k_a[S_0]} = \frac{1}{k_a[S_0]}\left(1 + \frac{k_q}{k_f}[Q]\right)$$

$$\frac{I_f^0}{I_f} = 1 + \frac{k_q}{k_f}[Q] \qquad (I_f^0 \equiv k_a[S_0])$$

(Stern-Volmer 식)



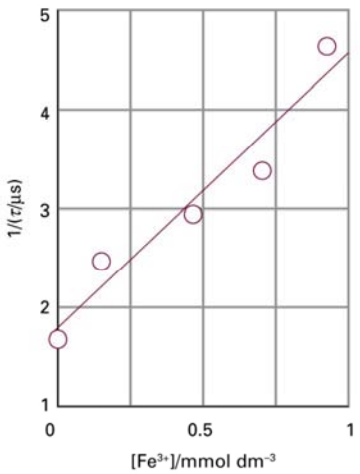
# 형광-소광 반응과 형광 수명 측정

- 펄스 광원에 의한 $k_f$ 와 $k_q$ 결정 방법 ($k_{ic}$와 $k_{isc}$의 값은 매우 작아 무시한다고 가정)

$$\frac{d[S_1]}{dt} = -k_f[S_1] - k_q[Q][S_1] = -(k_f + k_q[Q])[S_1]$$

$$\Rightarrow [S_1] = [S_1]_0 e^{-t/\tau} \qquad \left(\tau = \frac{1}{k_f + k_q[Q]}\right)$$

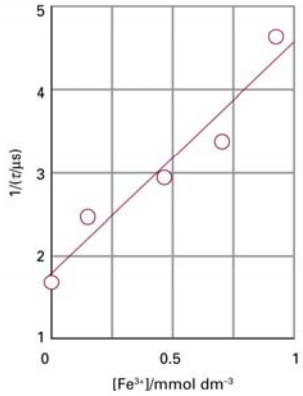
$$\tau = \frac{1}{k_f + k_q[Q]} \Rightarrow \frac{1}{\tau} = k_f + k_q[Q]$$





**(예제)** 수용액에서 용해된 $O_2$에 의한 tryptophan 형광의 소광 반응을 348 nm에서 방출되는 형광 수명을 측정함으로서 관측하였다. 얻어진 자료가 다음과 같을 때 소광 속도상수를 구하여라.

| $[O_2]$ / ($10^{-2}$ mol $dm^{-3}$) | $\tau$/($10^{-9}$ s) |
|---|---|
| 0 | 2.6 |
| 2.3 | 1.5 |
| 5.5 | 0.92 |
| 8.0 | 0.71 |
| 10.8 | 0.57 |

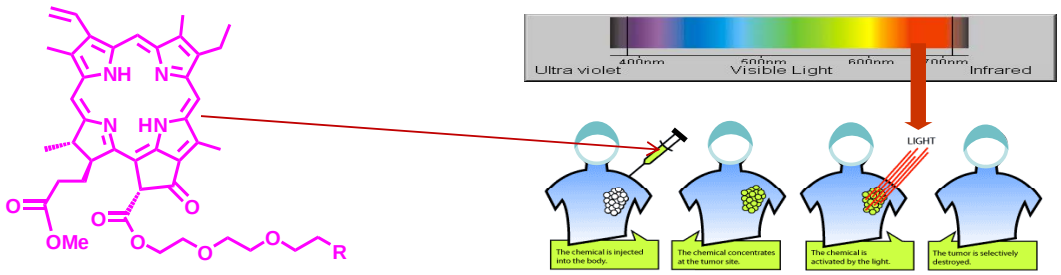





## 4.3 광역학 치료법

- 광역학 치료법 (Phtodynamic theraphy, PDT): 광감각제 (photosensitizer)를 이용하여 반응성이 매우 좋은 단일항 산소를 생성하고, 이를 이용하여 표적 세포의 기능을 파괴하는 치료법

$$\begin{aligned}
\text{Absorption:} &\quad {}^1P + h\nu \longrightarrow {}^1P^* \\
\text{Intersystem crossing:} &\quad {}^1P^* \longrightarrow {}^3P^* \\
\text{Photosensitization:} &\quad {}^3P^* + {}^3O_2 \longrightarrow P + {}^1O_2{}^* \\
\text{Oxidation reaction:} &\quad {}^1O_2{}^* + \text{reactants} \longrightarrow \text{products}
\end{aligned}$$





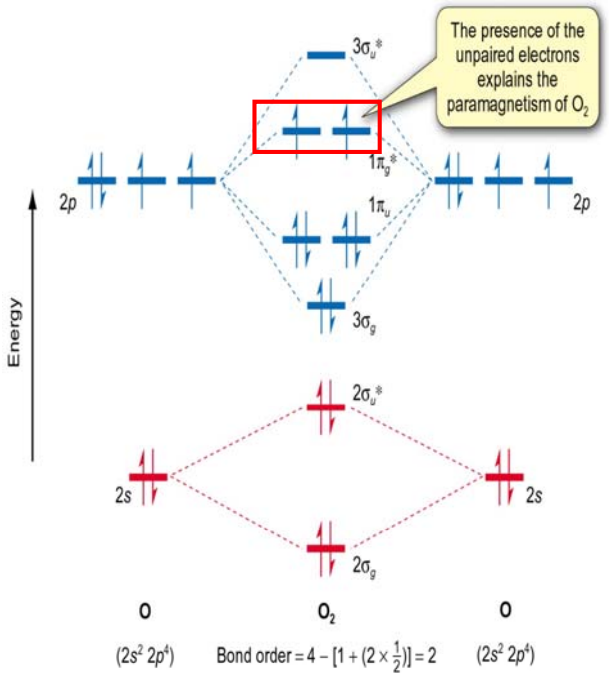
# $O_2$는 상자기성인가 반자기성일까?

Lewis 구조식에 의하면 $O_2$는 짝짓지 않은 전자를 가지고 있지 않음. → **반자기성 (?)**





$O_2$ 가 상자기성임을 보여주는 실험



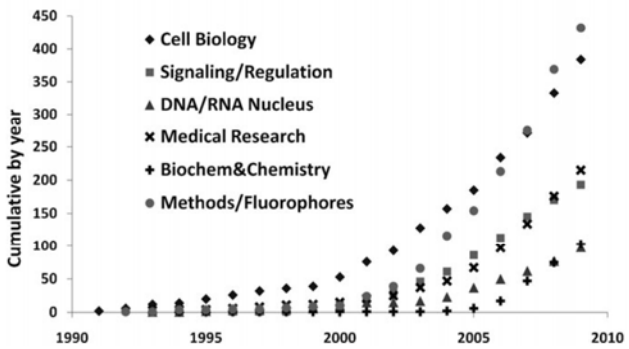



## 4.4 Fluorescence resonance energy transfer (FRET)

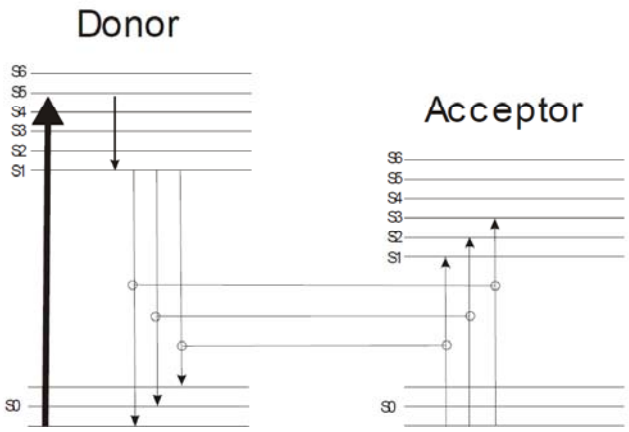
- Fluorescence resonance energy transfer (=Förster resonance energy transfer, FRET): 두 개의 형광 분자 사이에 짧은 거리 (10 nm 미만)에서 일어나는 에너지 이동 현상.

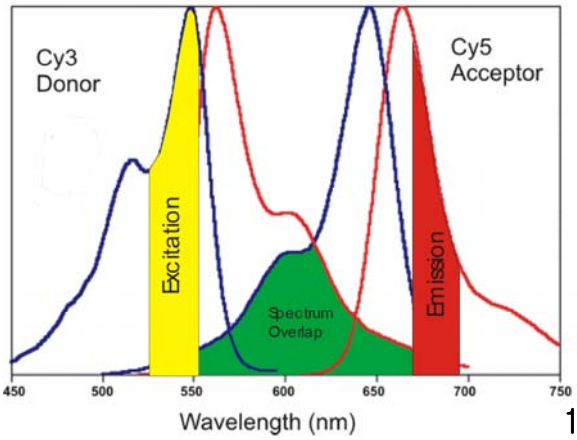



- 1946년에 Theodor Förster에 의해 FRET 현상이 발표 됨.
- Cumulative number of FRET publications by year in broad categories



# FRET mechanism



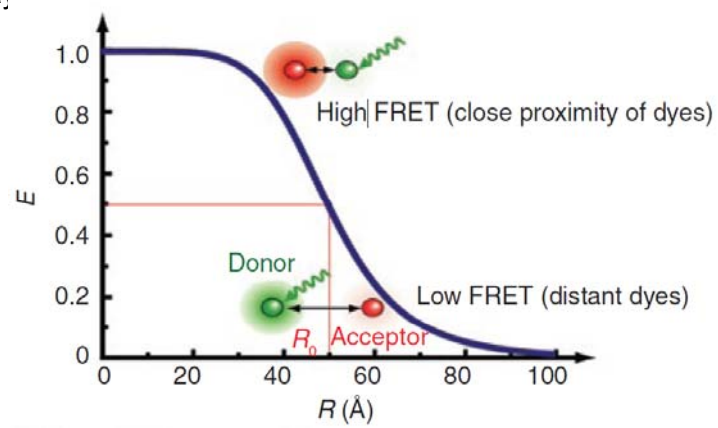



# FRET efficiency

- FRET efficiency ($E_T$)

$$E_T = 1 - \frac{\Phi_f^{\mathrm{D}}}{\Phi_{f,0}^{\mathrm{D}}} = \frac{k_{\mathrm{FRET}}}{k_{\mathrm{FRET}} + k_R + k_{\mathrm{NR}}} = \frac{1}{1 + (k_R + k_{\mathrm{NR}})/k_{\mathrm{FRET}}} = \frac{1}{1 + (R/R_0)^6}$$

  - $k_{FRET}$: FRET transfer rate
  - $k_R$: radiative decay rate
  - $k_{NR}$: nonradiative decay
  - $R_0$: Förster distance



# Förster distance

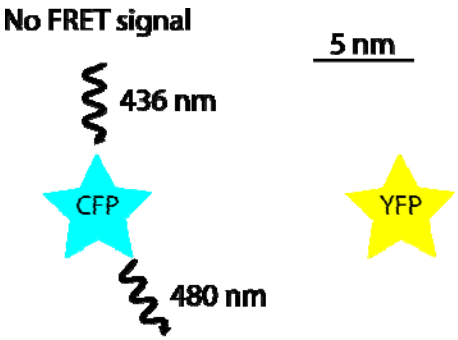
**Table 1.** Förster distances $R_0$ and photophysical properties for FP-based FRET pairs.

| Donor–acceptor[a] | $R_0$[b] [nm] | FP photophysical properties[c] FP | QY | EC | Ex. | Em. |
|---|---|---|---|---|---|---|
| CFP–YFP | 4.755 | CFP | 0.4 | 32.5 | 433 | 475 |
| Cerulean–Venus | 5.215 | Cerulean | 0.62 | 43 | 433 | 475 |
| mTFP–Venus | 5.701 | mTFP | 0.85 | 64 | 462 | 492 |
| mTFP–mKO2 | 5.29 | GFP | 0.6 | 56 | 488 | 507 |
| mTFP–tdTomato | 6.201 | YFP | 0.61 | 83.4 | 514 | 527 |
| mTFP–mTagRFP | 5.058 | Venus | 0.57 | 92.2 | 515 | 528 |
| mTFP–mKate | 4.559 | mOrange | 0.69 | 71 | 548 | 562 |
| GFP–mOrange | 5.381 | mKO2 | 0.62 | 63.8 | 551 | 565 |
| GFP–mCherry | 5.119 | tdTomato | 0.69 | 138 | 554 | 581 |
| Venus–tdTomato | 6.354 | mTagRFP | 0.41 | 81 | 558 | 584 |
| Venus–mCherry | 5.52 | mCherry | 0.22 | 72 | 587 | 610 |
| Venus–mKate | 5.042 | mKate | 0.33 | 45 | 589 | 636 |

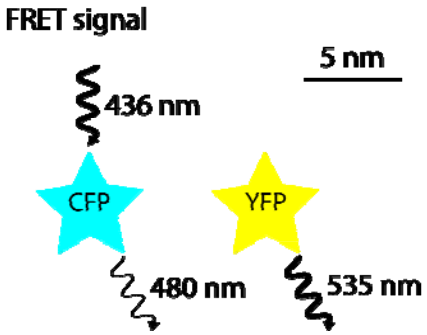


# Cyan fluorescence protein (CFP) – yellow fluorescence protein (YFP) FRET

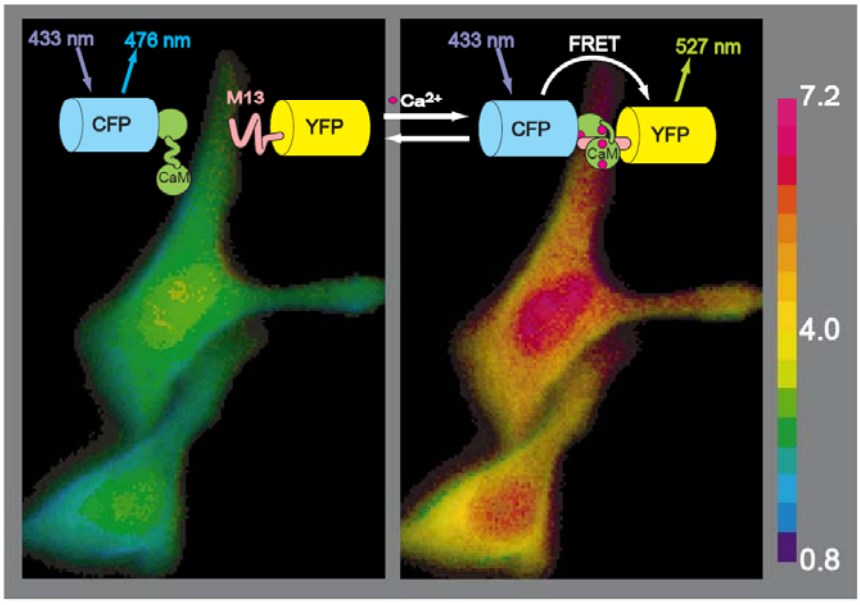
• CFP와 YFP의 거리가 5 nm 이상일 때




• CFP와 YFP의 거리가 5 nm 이하일 때



# FRET induced by $Ca^{2+}$ binding to calmodulin




Cyan fluorescence protein (CFP) labeled calmodulin and yellow fluorescence protein (YFP) labeled calmodulin binding peptide (M13–YFP) were coexpressed. High $Ca^{2+}$ levels (right) lead to binding and FRET emission of YFP (pseudo color red); low $Ca^{2+}$ levels (left) lead to little FRET and mostly blue emission (pseudocolor green).

Tsien & Miyawaki, Science **280**, 1954–1955 (1998).



# Chapman model for ozone formation

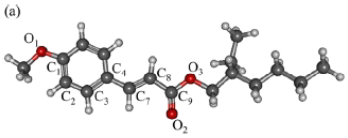
- 지구 대기권

- 오존 형성에 대한 Chapman 모델

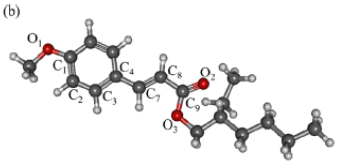
| | |
|---|---|
| Absorption: | $^1P + h\nu \longrightarrow {}^1P^*$ |
| Intersystem crossing: | $^1P^* \longrightarrow {}^3P^*$ |
| Photosensitization: | $^3P^* + {}^3O_2 \longrightarrow P + {}^1O_2{}^*$ |
| Oxidation reaction: | $^1O_2{}^* + \text{reactants} \longrightarrow \text{products}$ |

# Photochemistry for sunscreens

- 2-ethylhexyl-4-methoxycinnamate (EHMC): 자외선 차단제

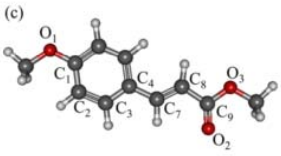


s-cis-EHMC

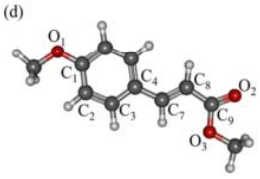


s-trans-EHMC

- methyl-4-methoxycinnamate (MMC): EHMC의 핵심 부분

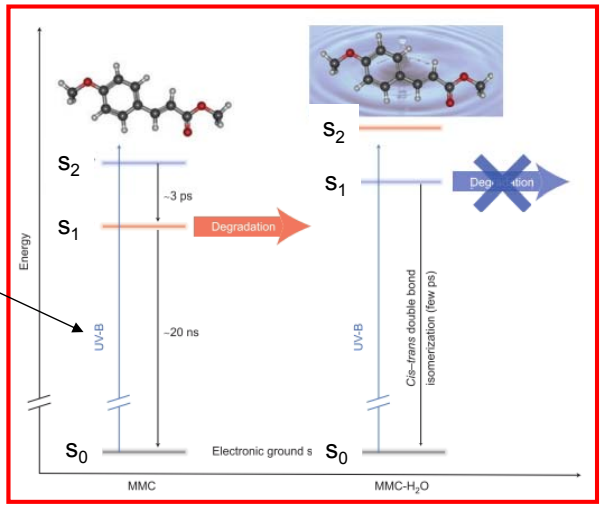


s-cis-MMC



s-trans-MMC

290-320 nm




J. Phys. Chem. Lett. **5**, 2464 (2014), Nat. Chem. **6**, 955 (2014)